EECS 242 Assignment 1 Fall 2025

Due at **11:59PM September 18, 2025**

What to submit: Submit a PDF to CatCourses. You can use the provided .tex and put your answers in solution sections bellow. Select all choices that apply for multiple-choice problems. You should also briefly show how solutions for multiple-choice problems are derived.

Student Name: **Po-Yun Cheng (276296347)**

Part I: Linear Algebra

1. Which of the following vectors are in the span of the vectors $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$?

   (a) $\begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}$

   (b) $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

   (c) $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

   (d) $\begin{bmatrix} 7 \\ -1 \\ 8 \end{bmatrix}$

   Solution:

   To see if a vector w is in the span of $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$, we verify if

   $$a(1, 2, -1) + b(3, 1, 2) = w$$

   has a solution. Therefore we have the solution as **(a), (d)**

2. Let

   $$A = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

   Find the inverse of AB^T.

   Solution: $AB^T = \begin{bmatrix} 5 & 0 \\ 6 & 2 \end{bmatrix}$ then $(AB^T)^{-1} = \frac{1}{10} \begin{bmatrix} 2 & 0 \\ -6 & 5 \end{bmatrix} = \begin{bmatrix} 0.2 & 0 \\ -0.6 & 0.5 \end{bmatrix}$

3. Which of the following statement is always true?

   (a) If A is a 3x5 matrix and B is a 5x4 matrix, then $(AB)^T$ is a 3x4 matrix.

   (b) If $A = A^T$, then the diagonal entries of A must be either 0 or 1s.

   (c) If $AB = A^T B^T$, then A and B must be of the same size.

   (d) $AA^T = A^T A$

   Solution:

   (a): false, it's 4x3.
   (b): false, it's symmetric, but not restrict to 0 or 1.
   (c): false, let's say A is $n \times m$, B is $m \times k$. Since A^T and B^T can be multiply, we have $n = k$. However, $n \times m$ not equal to $m \times n$ unless $n = m$.
   (d): false.
   None of them are true.

4. Show that the following vectors form a linearly dependent set in $\mathbb{R}^4$ by expressing $\boldsymbol{v}_2$ as a linear combination of the other two.

$$\boldsymbol{v}_1 = \begin{bmatrix} 6 \\ 0 \\ 5 \\ 1 \end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix} 0 \\ 3 \\ 1 \\ -1 \end{bmatrix}, \boldsymbol{v}_3 = \begin{bmatrix} 4 \\ -7 \\ 1 \\ 3 \end{bmatrix}$$

   Solution:

$$v_2 = \frac{2}{7}v_1 - \frac{3}{7}v_3$$

Part II: Probability

5. Suppose that $P(A \cap B) = 0.4$ and $P(B) = 0.9$. Find $P(A|B)$.

   Solution:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{4}{9}$$

6. A random variable, X, has the probability distribution table as shown.

x	-2	-1	0	1	2
$P(X = x)$			0.4	0.1	0.1

   Assume that $P(X = -2) = P(X = -1))$. Compute the expectation and variance of X.

   Solution:

   $P(X = -2) = P(X = -1) = 0.2$
   expectation: $E[X] = -0.4 - 0.2 + 0 + 0.1 + 0.2 = -0.3$
   variance: $Var(X) = E[X^2] - (E[X])^2 = 1.5 - 0.09 = 1.41$

7. A motor insurance company insures drivers in age group A, B and C. 40% of the customers are in group A, 25% are in B, and 35% are in group C. The companys record shows that each year, 2% of customers in age group A, 1% in group B and 1.5% in group C made a claim. Given that a driver made a claim, what is the probability that the driver is from age group C?

Solution:

$$\frac{0.35 * 0.015}{0.4 * 0.02 + 0.25 * 0.01 + 0.35 * 0.015} = \frac{0.00525}{0.01575} = \frac{1}{3}$$

Part III: Neural Network

8. Which of the followings would you consider to be valid activation functions?

   (a) $f(x) = -min(2, x)$

   (b) $f(x) = 0.9x + 1$

   (c) $f(x) = \begin{cases} min(x, 0.1x) \text{ if } x \geq 0 \\ min(x, 0.1x) \text{ if } x < 0 \end{cases}$

   (d) $f(x) = \begin{cases} max(x, 0.1x) \text{ if } x \geq 0 \\ min(x, 0.1x) \text{ if } x < 0 \end{cases}$

   Solution:

   a valid activation function should be non-linear and differentialbe.
   (a) is is like ReLU.
   (b) is not valid as it's linear.
   (c) is like leaky ReLU.
   (d) is basically $f(x) = x$ (identity function), I saw some resources state that it could be activation function, while I don't think so.
   answer: **(a), (c)**

9. Which of the following indicates overfitting?

   (a) High training error, high test error

   (b) Low training error, low test error

   (c) Low training error, high test error

   (d) High training error, low test error

   Solution:

   (c) because the model is "overfitting" on the training data.

10. Suppose we are training a simple neural network with two layers for regression. The network takes two-dimensional input $[x_1, x_2]$ and gives a scalar $\bar{y}$. The first layer is a linear layer followed by ReLU. The second layer is also a linear layer followed by ReLU.

First layer: $h_1 = ReLU(ax_1 + bx_2 + c)$

Second layer: $\bar{y} = ReLU(dh_1 + e)$

Loss: $L = (y - \bar{y})^2$

The network parameters are initialized as follows,

$$a = 2, b = 3, c = 1, d = 2, e = -5 \tag{1}$$

(a) Given one training data point $[x_1, x_2] = [1, 0]$ and its ground truth $y = 3$, compute h_1,$\bar{y}$,L in the forward pass.

(b) Compute the gradient of the loss w.r.t. network paraters a,b,c,d,e respectively.

(c) If the learning rate is set as 0.1, compute the updated value of a,b,c,d,and e after one iteration of gradient descent.

Solution:

(a)

$$\begin{aligned} h_1 &= \max(2 \times 1 + 3 \times 0 + 1, 0) = 3 \\ \bar{y} &= \max(2 \times 3 - 5, 0) = 1 \\ L &= (3 - 1)^2 = 4 \end{aligned}$$

(b)

$$\begin{aligned} \frac{\partial L}{\partial e} &= \frac{\partial L}{\partial \bar{y}} \frac{\partial \bar{y}}{\partial e} = 2(\bar{y} - y) \times 1 = -4 \\ \frac{\partial L}{\partial d} &= \frac{\partial L}{\partial \bar{y}} \frac{\partial \bar{y}}{\partial d} = 2(\bar{y} - y) \times h_1 = -12 \\ \frac{\partial L}{\partial c} &= \frac{\partial L}{\partial \bar{y}} \frac{\partial \bar{y}}{\partial h_1} \frac{\partial h_1}{\partial c} = 2(\bar{y} - y) \times d \times 1 = -8 \\ \frac{\partial L}{\partial b} &= \frac{\partial L}{\partial \bar{y}} \frac{\partial \bar{y}}{\partial h_1} \frac{\partial h_1}{\partial b} = 2(\bar{y} - y) \times d \times x_2 = 0 \\ \frac{\partial L}{\partial a} &= \frac{\partial L}{\partial \bar{y}} \frac{\partial \bar{y}}{\partial h_1} \frac{\partial h_1}{\partial a} = 2(\bar{y} - y) \times d \times x_1 = -8 \end{aligned}$$

(c)

$$\begin{aligned} a' &= 2 - 0.1 \times (-8) = 2.8 \\ b' &= 3 - 0.1 \times 0 = 3 \\ c' &= 1 - 0.1 \times (-8) = 1.8 \\ d' &= 2 - 0.1 \times (-12) = 3.2 \\ e' &= -5 - 0.1 \times (-4) = -4.6 \end{aligned}$$